

## Solution Code - Module 3-Core Java - PatternMatchingSwitch

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Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

### Source File: Q30\_PatternMatchingSwitch.java

```
1 | public class Q30_PatternMatchingSwitch {
2 |     // Method that accepts Object and uses pattern matching in switch
3 |     public static String getObjectTypeInfo(Object obj) {
4 |         return switch (obj) {
5 |             case null -> "The object is null.";
6 |             case Integer i -> "The object is an Integer: " + i + " (Value squared: " + (i * i) + ")";
7 |             case String s -> "The object is a String: \"" + s + "\"" (Length: " + s.length() + ")";
8 |             case Double d -> "The object is a Double: " + d + " (Value halved: " + (d / 2.0) + ")";
9 |             case Boolean b -> "The object is a Boolean: " + b + " (Inverse: " + (!b) + ")";
10 |             default -> "The object is of an unknown type: " + obj.getClass().getSimpleName();
11 |         };
12 |     }
13 |
14 |     public static void main(String[] args) {
15 |         System.out.println("=== Pattern Matching for switch (Java 21) ===");
16 |
17 |         Object[] testObjects = {
18 |             10,
19 |             "Hello, Java 21!",
20 |             3.14159,
21 |             true,
22 |             new Object(),
23 |             null
24 |         };
25 |
26 |         for (Object obj : testObjects) {
27 |             System.out.println("Testing object: " + (obj == null ? "null" : obj));
28 |             System.out.println("Result: " + getObjectTypeInfo(obj));
29 |             System.out.println();
30 |         }
31 |     }
32 | }
```